

A logarithmic strip tail destroys every finite-order Laurent decomposition at an internal tangency

Candidate full negative answer to Question 34 of arXiv:1605.08289

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Status: candidate counterexample, likely valid; full for every finite p in the source's parameter set. For each integer $p \geq 0$ and every round disc internally tangent to the unit disc, an explicit $F_p \in A^p(\Omega)$ is constructed which is not in $A^p(\mathbb{D}) + A_0^p(\widehat{\mathbb{C}} \setminus \overline{D'})$. The construction turns the cusp into a vertical strip and leaves a one-sided logarithmic tail in the p -th derivative.

Source question

Let D' be a disc contained in $\mathbb{D}$ whose boundary is tangent to $\mathbb{T}$ at 1, and put $\Omega = \mathbb{D} \setminus \overline{D'}$. The source proves the Laurent decomposition for $p = \infty$ and then asks:

For $p < +\infty$, is there an $F \in A^p(\Omega)$ that cannot be written as the sum of a function in $A^p(\mathbb{D})$ and another in $A_0^p(\widehat{\mathbb{C}} \setminus \overline{D'})$?

This is Question 34 on PDF page 13 of Georgakopoulos's paper [1, Question 34]. The page containing the question and its Cauchy transform criterion is reproduced below.

for all triples $x, y, z \in A$, $x \neq z$. Clearly, diffeomorphisms $\mathbb{T} \rightarrow \mathbb{T}$ are quasi-symmetric.

5. INTERNALLY TANGENT CIRCLES

Take a disc D' in the interior of D , with $\partial D'$ tangent at 1 to $\mathbb{T} = \partial D$, and let $\Omega = D - \overline{D'}$. We denote by $\mathbb{T}$ the boundary of D (as usual) and the boundary of D' by γ . We examine whether or not every function $f \in A^p(\Omega)$ has a decomposition as $f = g + h$ for $g \in A^p(D)$ and $h \in A_0^p(\hat{\mathbb{C}} \setminus \overline{D'})$. The case $p = +\infty$ is easily dealt with:

Theorem 32. *Every function in $A^\infty(\Omega)$ can be written uniquely as the sum of a function in $A^\infty(D)$ and of a function in $A_0^\infty(\hat{\mathbb{C}} \setminus \overline{D'})$.*

Proof. If $f \in A^\infty(\Omega)$ then clearly, $f \in C^\infty(\mathbb{T} - \{1\})$. The derivatives of f extend continuously over $\mathbb{T}$ hence $f \in C^\infty(\mathbb{T})$; we similarly have that $f \in C^\infty(\partial D')$. By Theorem 6, there are $g \in A^\infty(D)$ and $h \in A_0^\infty(\hat{\mathbb{C}} \setminus \overline{D})$ so that $f = g + h$ on the unit circle. We extend h on D^c by setting $h = f - g$ over $D - D'$; the derivatives of h are also continuously extended this way. It is easy to see that the extended h is holomorphic over $\overline{D'}^c$, so $h \in A_0^\infty(\hat{\mathbb{C}} \setminus \overline{D'})$ as desired. This proves the existence of the decomposition. For the uniqueness of this decomposition, we observe that Liouville's Theorem implies that $A^\infty(D) \cap A_0^\infty(\hat{\mathbb{C}} \setminus \overline{D'}) = 0$. $\square$

Combining this with Proposition 2 we obtain:

Corollary 33. $A^\infty(\Omega) = A^\infty(D) \oplus A_0^\infty(\hat{\mathbb{C}} \setminus \overline{D'})$.

Question 34. For $p < +\infty$, is there an $F \in A^p(\Omega)$ that can't be written as the sum of a function in $A^p(D)$ and another in $A_0^p(\hat{\mathbb{C}} \setminus \overline{D'})$?

We suspect that the decomposition does not hold (just as in the case for the circle/Jordan curve) and will now collect some sufficient conditions for a function $F \in A^p(\Omega)$ to not be in $A^p(D) + A_0^p(\hat{\mathbb{C}} \setminus \overline{D'})$.

The Cauchy transform is defined as

$$C_{\mathbb{T}}(F) = \frac{1}{2\pi i} \int_{\mathbb{T}} \frac{f(\zeta)}{\zeta - z} d\zeta$$

Assume $F = f + g$, $f \in A^p(D)$ and $g \in A_0^p(\hat{\mathbb{C}} \setminus \overline{D'})$. Then $C_{\mathbb{T}}(F) = C_{\mathbb{T}}(f) + C_{\mathbb{T}}(g)$. We calculate

$$C_{\mathbb{T}}(F) = \begin{cases} f(z) & \text{if, } z \in D \\ -g(z) & \text{if, } z \notin D \end{cases}$$

If γ is the boundary of $\partial D'$, we can similarly prove that

$$C_{\gamma}(F) = \begin{cases} f(z) & \text{if, } z \in D' \\ -g(z) & \text{if, } z \notin D' \end{cases}$$

We have

$$\lim_{z \rightarrow 1, z \in D} C_{\mathbb{T}}(F) = \lim_{z \rightarrow 1, z \in \Omega} C_{\mathbb{T}}(F) = \lim_{z \rightarrow 1, z \in D'} C_{\gamma}(F) \in \mathbb{C}$$

and

$$\lim_{z \rightarrow 1, z \notin D} C_{\mathbb{T}}(F) = \lim_{z \rightarrow 1, z \notin D} C_{\gamma}(F) = \lim_{z \rightarrow 1, z \notin \Omega} C_{\gamma}(F) \in \mathbb{C}$$

Here $p \in \{0, 1, 2, \dots\}$, as in the source's definition of A^p .

Proof intuition

Inversion at the tangency point, $w = (1 - z)^{-1}$, sends the crescent Ω to a vertical strip. Its two infinite ends both represent the same boundary point $z = 1$. An analytic cutoff on the strip can tend to 1 at

one end and to 0 exponentially at the other, while the resulting function still has a single limit at the tangency point.

We multiply that cutoff by $w^{-p}/\text{Log}(2-iw)$. The factor w^{-p} absorbs the p differentiations in the z variable; after p derivatives the positive end still has size $1/\log|w|$, whereas the negative end is exponentially small. On the outer circle this becomes a one-sided boundary tail $1/\log(1/t)$. Its Cauchy projection grows like $\log\log(1/(1-\rho))$ as $\rho \uparrow 1$. If an A^p Laurent decomposition existed, differentiating it p times would make this projection an element of $A(\mathbb{D})$, which must have a finite limit at 1.

The counterexample

Write

$$D' = D'_r := \{z : |z - (1-r)| < r\}, \quad 0 < r < 1,$$

which is the general disc internally tangent to $\mathbb{T}$ at 1. Set

$$a = \frac{1}{2}, \quad b = \frac{1}{2r}, \quad L = b - a, \quad W(z) = \frac{1}{1-z}.$$

For $a < \Re w < b$, define

$$\begin{aligned} E(w) &= \exp\left(\frac{\pi i(w-a)}{L}\right), & \chi(w) &= \frac{1}{1-iE(w)}, \\ \Lambda(w) &= \text{Log}(2-iw), & \Phi_p(w) &= \frac{w^{-p}\chi(w)}{\Lambda(w)}, \end{aligned}$$

where Log is the principal logarithm. Our function is

$$F_p(z) = \Phi_p(W(z)), \quad z \in \Omega. \tag{1}$$

Theorem 1. *For every integer $p \geq 0$, the function F_p in (1) belongs to $A^p(\Omega)$ but*

$$F_p \notin A^p(\mathbb{D}) + A_0^p(\widehat{\mathbb{C}} \setminus \overline{D'_r}).$$

Consequently Question 34 has an affirmative answer for every finite p .

Proof

1. The strip and its two ends

For $w = W(z)$ one has

$$|z| < 1 \iff \Re w > \frac{1}{2}, \quad |z - (1-r)| > r \iff \Re w < \frac{1}{2r}.$$

Thus W maps Ω conformally onto the strip $S = \{a < \Re w < b\}$. The two limits $\Im w \rightarrow +\infty$ and $\Im w \rightarrow -\infty$ both correspond to $z \rightarrow 1$ through the two arms of the cusp.

The map E sends S to the upper half-plane. Uniformly for $a \leq \Re w \leq b$,

$$\chi(w) = 1 + O(e^{-\pi \Im w / L}) \quad (\Im w \rightarrow +\infty), \quad \chi(w) = O(e^{-\pi |\Im w| / L}) \quad (\Im w \rightarrow -\infty), \tag{2}$$

and the same estimates, up to polynomial factors, hold after any fixed number of w -derivatives. Indeed, the only zero of $1 - i\zeta$ is $\zeta = -i$, outside the closed upper half-plane.

Moreover $2 - iw$ lies in the open lower half-plane throughout the closed strip, because $\Im(2 - iw) = -\Re w < 0$. Hence Λ is holomorphic on a neighborhood of every finite point of the closed strip and never vanishes there. At either end,

$$|\Lambda(w)| \sim \log|w|. \tag{3}$$

2. Finite-order regularity at the cusp

Since $dW/dz = W^2$, differentiation in z becomes

$$\mathcal{T} := w^2 \frac{d}{dw}.$$

Repeated product differentiation using (2)–(3) gives, for $0 \leq k \leq p$ and $\Im w \rightarrow +\infty$,

$$\mathcal{T}^k \Phi_p(w) = (-1)^k \frac{p!}{(p-k)!} \frac{w^{k-p}}{\Lambda(w)} + O\left(\frac{|w|^{k-p}}{|\Lambda(w)|^2}\right) + o(1), \quad (4)$$

uniformly on the closed strip. Terms in which a derivative hits χ are exponentially small; a derivative hitting Λ^{-1} gains one power of w but also one additional inverse power of Λ . This also proves (4) by induction. At the negative end,

$$\mathcal{T}^k \Phi_p(w) = O(e^{-c|\Im w|} |w|^{N_k}) = o(1) \quad (5)$$

for suitable $c > 0$ and N_k .

Equations (4) and (5) show that $\mathcal{T}^k \Phi_p(w) \rightarrow 0$ at both ends for every $k \leq p$ (also when $k = p$, because $|\Lambda(w)| \rightarrow \infty$). Away from the cusp all functions are holomorphic on a neighborhood of the two boundary arcs. Therefore each $F_p^{(k)}$, $0 \leq k \leq p$, extends continuously to $\bar{\Omega}$ by the value 0 at 1. This proves

$$F_p \in A^p(\Omega). \quad (6)$$

For later use, put $Q_p = F_p^{(p)}$ and $A_p = (-1)^p p!$. The case $k = p$ of (4) says

$$Q_p(z) = \frac{A_p}{\Lambda(W(z))} + O\left(\frac{1}{|\Lambda(W(z))|^2}\right) \quad (\Im W(z) \rightarrow +\infty), \quad (7)$$

while Q_p is exponentially small, up to a polynomial factor, at the other end.

3. The outer Cauchy projection diverges

On the outer circle, with $z = e^{it}$,

$$W(e^{it}) = \frac{1}{2} + \frac{i}{2} \cot \frac{t}{2}. \quad (8)$$

Thus (7) yields, as $t \downarrow 0$,

$$\frac{Q_p(e^{it})}{A_p} = \frac{1}{\log(1/t)} + O\left(\frac{1}{\log^2(1/t)}\right), \quad (9)$$

whereas for $t \uparrow 0$ from below,

$$Q_p(e^{it}) = O(|t|^{-N} e^{-c/|t|}) \quad (10)$$

for some $c, N > 0$.

For $0 < \rho < 1$, let

$$\mathcal{C}_p(\rho) = \frac{1}{2\pi i} \int_{\mathbb{T}} \frac{Q_p(\zeta)}{\zeta - \rho} d\zeta = \frac{1}{2\pi} \int_{-\pi}^{\pi} Q_p(e^{it}) \frac{e^{it}}{e^{it} - \rho} dt.$$

The exact imaginary part of the kernel is

$$\Im \frac{e^{it}}{e^{it} - \rho} = -\frac{\rho \sin t}{(1 - \rho)^2 + 2\rho(1 - \cos t)}. \quad (11)$$

Put $\delta = 1 - \rho$. On $4\delta \leq t \leq \varepsilon$, the right side of (11) is $-1/t$ up to an error whose integral against $1/\log(1/t)$ remains bounded. The real part of the kernel, multiplied by the imaginary error in (9), also has uniformly bounded integral. The interval $0 < t < 4\delta$ contributes $O(1/\log(1/\delta))$; the negative arc is uniformly bounded by (10); and arcs away from 1 are uniformly bounded. Consequently

$$\begin{aligned}\Im \frac{\mathcal{C}_p(\rho)}{A_p} &= -\frac{1}{2\pi} \int_{4\delta}^{\varepsilon} \frac{dt}{t \log(1/t)} + O(1) \\ &= -\frac{1}{2\pi} \log \log \frac{1}{\delta} + O(1) \longrightarrow -\infty.\end{aligned}\tag{12}$$

In particular, the outer Cauchy projection of Q_p has no finite limit at 1 from inside $\mathbb{D}$.

4. Contradiction to a Laurent decomposition

Suppose that $F_p = g + h$ with

$$g \in A^p(\mathbb{D}), \quad h \in A_0^p(\widehat{\mathbb{C}} \setminus \overline{D_r}).$$

Differentiating p times gives $Q_p = g^{(p)} + h^{(p)}$, with $g^{(p)} \in A(\mathbb{D})$. Since $h^{(p)}$ is holomorphic on the exterior of the unit circle and vanishes at infinity, its Cauchy integral over $\mathbb{T}$ is zero inside $\mathbb{D}$. Cauchy's formula therefore gives

$$\mathcal{C}_p(z) = g^{(p)}(z), \quad z \in \mathbb{D}.$$

The right side extends continuously to $z = 1$, contradicting (12). This proves Theorem 1. $\square$

Clarification of the source's final sentence

The source later asks, “in particular,” whether some $F \in A(\mathbb{D})$ can have a divergent inner limit of its outer-circle Cauchy transform. Literally this subquestion has the immediate negative answer

$$C_{\mathbb{T}}(F)(z) = F(z) \longrightarrow F(1), \quad z \rightarrow 1, z \in \mathbb{D},$$

by Cauchy's formula and the definition of the disc algebra. The substantive Question 34 concerns $F \in A^p(\Omega)$; Theorem 1 answers that question fully. For $p > 0$ the proof uses the Cauchy projection of the p -th derivative, rather than claiming that the projection of F_p itself diverges.

Verification and novelty bounds

The strip identities, the two endpoint asymptotics, and the Cauchy-kernel estimate are symbolic. The included script `code/check_asymptotics.py` independently differentiates the formula for $p = 0, 1, 2, 3$ and checks the one-sided boundary ratio and the predicted log–log growth numerically; it is a sanity check, not part of the proof.

A bounded novelty search on 11 August 2026 covered the run's registry, solution, attempt, and proof-gap indexes; the exact arXiv id and title; “Question 34” with Laurent/ A^p keywords; the author name; and “internally tangent circles” with the decomposition and Cauchy-transform keywords. It found the source paper and later work on A^p boundary traces for Jordan domains, but no later answer to this tangent-disc question or this strip/logarithmic construction. Novelty confidence is therefore moderate, not definitive.

Human review should focus on the induction in (4), uniformity at both strip ends, and the bounded-error reduction from (11) to (12). No numerical statement is used in the proof.

References

- [1] N. Georgakopoulos, *Extensions of the Laurent Decomposition and the spaces $A^p(\Omega)$* , arXiv:1605.08289 (2016).